

AI Multi-Agent Math Inquiries

A professional educational assistant that combines LangGraph orchestration, retrieval-augmented generation, reflection, and symbolic math tools.

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5 core agents

1150 indexed chunks

2 built-in tools

Presentation Roadmap

A structured walkthrough from the problem statement to verified project results.

1

Problem and motivation

2

Architecture, workflow, and
UML view

3

How the agents are
implemented

4

UI and explainability
features

5

Demo results and final
evaluation

Problem And Motivation

The project is designed to answer math questions with more structure than a single-pass chatbot.

Why The Problem Matters

- Math questions range from simple definitions to exact equation solving.
- Students need both explanation and accuracy, not only fluent text.
- Class demos work better when the system can show how its answer was built.

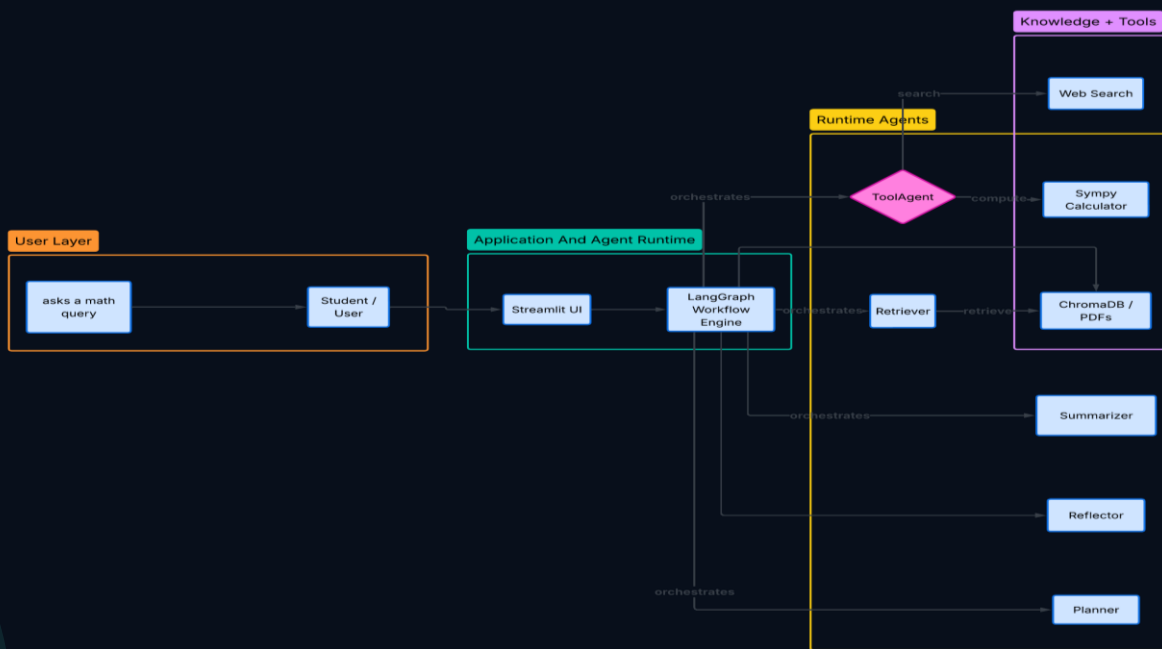
What This Project Adds

- A planner that classifies the query before acting.
- Evidence retrieval from local math documents.
- Answer checking through reflection metrics.
- Tool-based symbolic solving when precision matters.

ARCHITECTURE

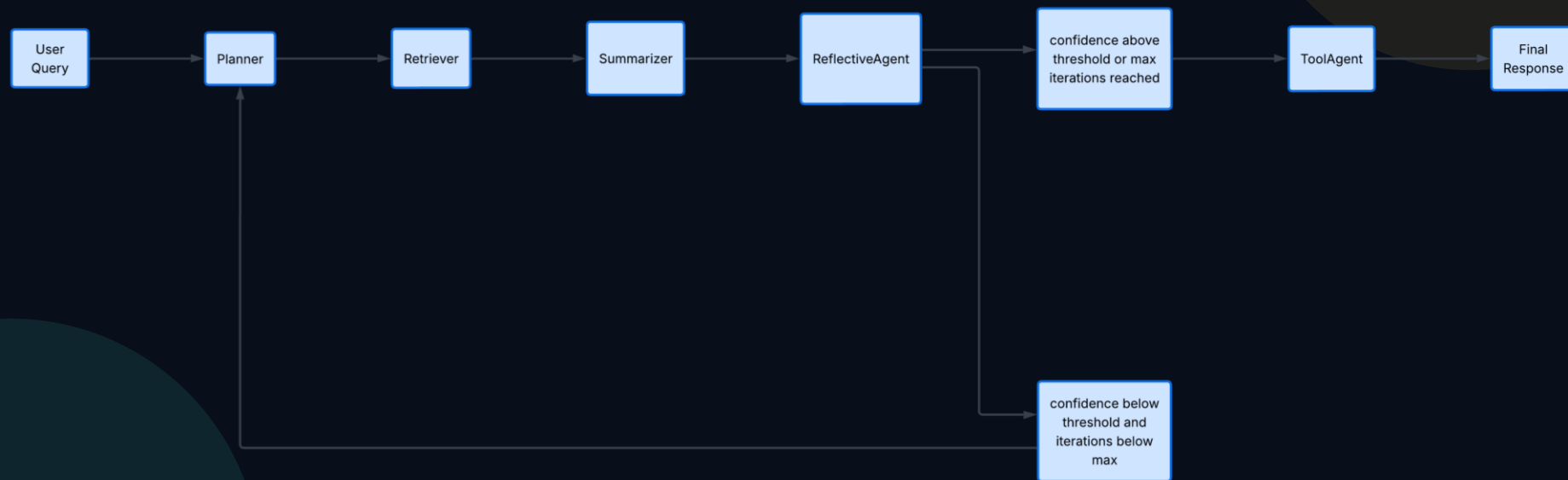
High-Level Architecture

This diagram shows how the user, UI, agent runtime, retrieval layer, and tool layer connect.



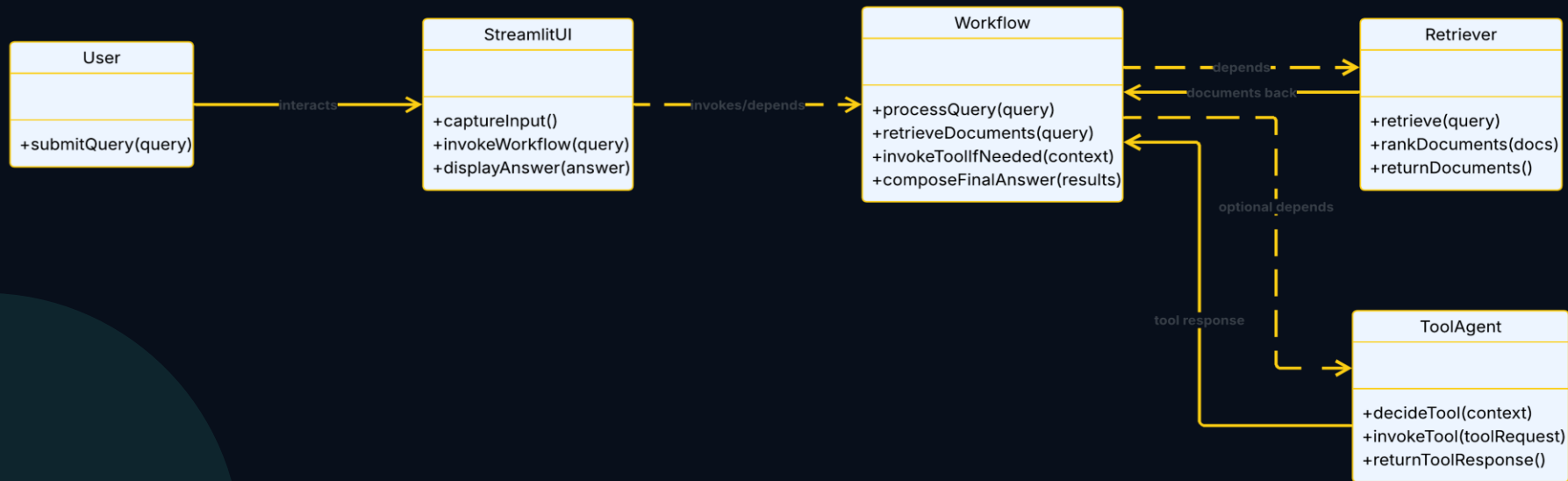
Workflow Diagram

This flowchart highlights the exact execution order and the retry loop driven by confidence.



UML View

A sequence-style view makes it easier to explain how the user request moves across the system.



Shared State And Control Flow

A single shared state object makes the workflow easier to extend, inspect, and debug.

Workflow State

```
user_query
retrieved_docs
summary
reflection_metrics
tool_results
iteration_count
metadata
```

Why This Design Works

- Every node reads from and writes to the same state object.
- Metadata captures sequence, timings, decisions, and trace details.
- The reflector can send the state back to the planner for another pass.

Planner Agent

The planner sets the direction of the workflow before retrieval and tool execution begin.

Planner Responsibilities

- Classify the query type: definition, calculation, proof, comparison, or research.
- Increase iteration count and react to reflection feedback.
- Choose retrieval breadth and suggest tools such as calculator or web search.

Code Highlight

```
query_type = _classify_query(query)
state["iteration_count"] += 1
confidence, next_action =
    _evaluate_feedback(state)
k_docs = _decide_retrieval_k(state)
selected_tools = _decide_tools(query,
    query_type)
```


Retriever, ChromaDB, And Ingestion

The retriever connects the question to the project knowledge base before answer generation begins.

Retriever Design

- Uses ChromaDB for persistent vector retrieval.
- Embeddings are generated with all-MiniLM-L6-v2.
- Ingestion loads JSONL and PDF files, then chunks and stores them.

Code Highlight

```
docs =  
self.retriever.retrieve(query,  
k=k)  
state["retrieved_docs"] = docs  
  
collection status:  
healthy  
document_count = 1150
```

Summarizer Agent

The summarizer tries to produce a grounded answer from retrieved evidence and falls back safely when needed.

Summarization Strategy

- Supports Mistral, OpenAI, or Gemini through configuration.
- Uses a prompt that restricts the answer to retrieved excerpts.
- Falls back to a local summarizer when the LLM path is unavailable.

Prompt Intent

Answer the user's question
using only the provided
document excerpts.

If the LLM path is unavailable,
use fallback summarization.

Reflective Agent And Retry Logic

Reflection gives the graph a quality checkpoint before the answer is finalized.

Reflection Metrics

- Scores factual correctness, completeness, and relevance.
- Computes a single confidence score for retry decisions.
- Provides feedback text and notes for diagnostics.

Confidence Formula

```
confidence =  
    0.45 * factual_correctness +  
    0.30 * completeness +  
    0.25 * relevance  
  
retry if confidence < 0.60
```

ToolAgent And Exact Math Output

The tool layer is where the system becomes more than a text summarizer.

Tool Layer

- The registry currently exposes a Sympy calculator and a web-search tool.
- Planner hints are applied first, then query heuristics add any missing tools.
- Calculator output can replace the summary when it is more authoritative.

Code Highlight

```
if item["tool"] ==  
"calculator":  
    state["summary"] =  
    result_text  
    state["tool_results"] =  
    None
```

```
Solve  $2x + 3 = 11$   
=> Solution:  $x = 4$ 
```

Frontend Experience And Diagnostics

The interface is designed to support both classroom presentation and technical inspection.

Chat Experience

- Streamlit chat interface for asking math questions.
- Conversation-style answers instead of raw console logs.
- Built to feel approachable in a live classroom demo.

Diagnostics

- Retrieved documents are visible after each run.
- Reflection scores and confidence are easy to inspect.
- Metadata and timings help explain system behavior.

Teaching Value

- Students can see not only the final answer but also how it was produced.
- The interface supports both demonstration and debugging.

RESULTS

Demo Result 1: Conceptual Query

A useful example of how retrieval quality affects the final explanation.

Question: What is the quadratic formula and when should I use it?

Observed Run

- The full agent sequence completed end to end.
- 5 documents were retrieved.
- Fallback confidence reached 70.8 percent.

What Worked

- The answer identified the quadratic formula idea correctly.
- The workflow remained explainable and traceable.

What To Improve

- The response mixed in unrelated PDF content.
- Concept-style retrieval still needs stronger filtering and ranking.

RESULTS

Demo Result 2: Symbolic Math Success

This example shows the strongest part of the project: exact output through tool augmentation.

Query: Solve $2x+y-z=3$; $4x-y+2z=1$; $-2x+3y+z=4$

Why This Result Matters

- The calculator tool was selected automatically.
- The summary was replaced by an exact symbolic result.
- The answer included matrix steps, not only the final values.

Observed Output

RREF:

$$\begin{bmatrix} 1 & 0 & 0 & 21/32 \\ 0 & 1 & 0 & 7/4 \\ 0 & 0 & 1 & 1/16 \end{bmatrix}$$

Solution: $x = 21/32$, $y = 7/4$, $z = 1/16$

CLOSE

Strengths, Limitations, And Next Steps

The project is already a strong prototype, and the next improvements are clear.

Strengths

- Clear multi-agent separation of responsibility.
- Explainable state, metadata, and diagnostics.
- Exact symbolic solving is a strong differentiator.

Limitations

- Concept-question retrieval can still be noisy.
- LLM quality depends on installed clients and API setup.
- Fallback scoring may feel more confident than the answer quality.

Next Steps

- Add reranking and smarter retrieval filtering.
- Cache embeddings for smoother classroom demos.
- Improve the integration of the modular UI layer.